

برنامج حساب الإحصائيات (#C)

Source Code Documentation

```
using System;
using System.Linq;
using System.Collections.Generic;

class Program
{
    static void Main()
    {
        List<int> data = new List<int> { 115, 182, 191, 31, 196,
1099, 5, 172, 10, 179, 83, 21, 20, 21, 186, 177, 195, 193, 188,
1 };

        data.Sort();
        int n = data.Count;

        // Mean
        double mean = data.Average();

        // Median
        double median = (n % 2 == 0) ? (data[n / 2 - 1] +
data[n / 2]) / 2.0 : data[n / 2];

        // Mode
        var mode = data.GroupBy(x => x)
                        .OrderByDescending(g => g.Count())
                        .First().Key;

        // Variance
        double variance = data.Sum(x => Math.Pow(x - mean, 2)) /
n;

        // Standard Deviation
        double stdDev = Math.Sqrt(variance);

        // Percentiles Function Logic
        double Percentile(double p)
        {
            double pos = (p / 100.0) * (n + 1);
```

```

        int index = (int)pos;

        if (index <= 0) return data[0];
        if (index >= n) return data[n - 1];

        double fraction = pos - index;
        return data[index - 1] + fraction * (data[index] -
data[index - 1]);
    }

    double P20 = Percentile(20);
    double P50 = Percentile(50);

    // Quartiles
    double Q1 = Percentile(25);
    double Q2 = median;
    double Q3 = Percentile(75);

    // Range
    int range = data.Max() - data.Min();

    // IQR
    double IQR = Q3 - Q1;

    // Summation
    int sum = data.Sum();

    // Output
    Console.WriteLine($"Mean = {mean}");
    Console.WriteLine($"Mode = {mode}");
    Console.WriteLine($"Median = {median}");
    Console.WriteLine($"Variance = {variance}");
    Console.WriteLine($"P20 = {P20}");
    Console.WriteLine($"P50 = {P50}");
    Console.WriteLine($"Q1 = {Q1}");
    Console.WriteLine($"Q2 = {Q2}");
    Console.WriteLine($"Q3 = {Q3}");
    Console.WriteLine($"Range = {range}");
    Console.WriteLine($"IQR = {IQR}");
    Console.WriteLine($"Standard Deviation = {stdDev}");
    Console.WriteLine($"Sum = {sum}");
}
}

```